Component Diagram for Keyboard driver

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Session:2023-2027

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Project Name: KEYBOAD DEVICE DRIVER

Session:2023-2027

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Subject: OPERATING SYSTEM

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Date of Submission: February 13, 2026

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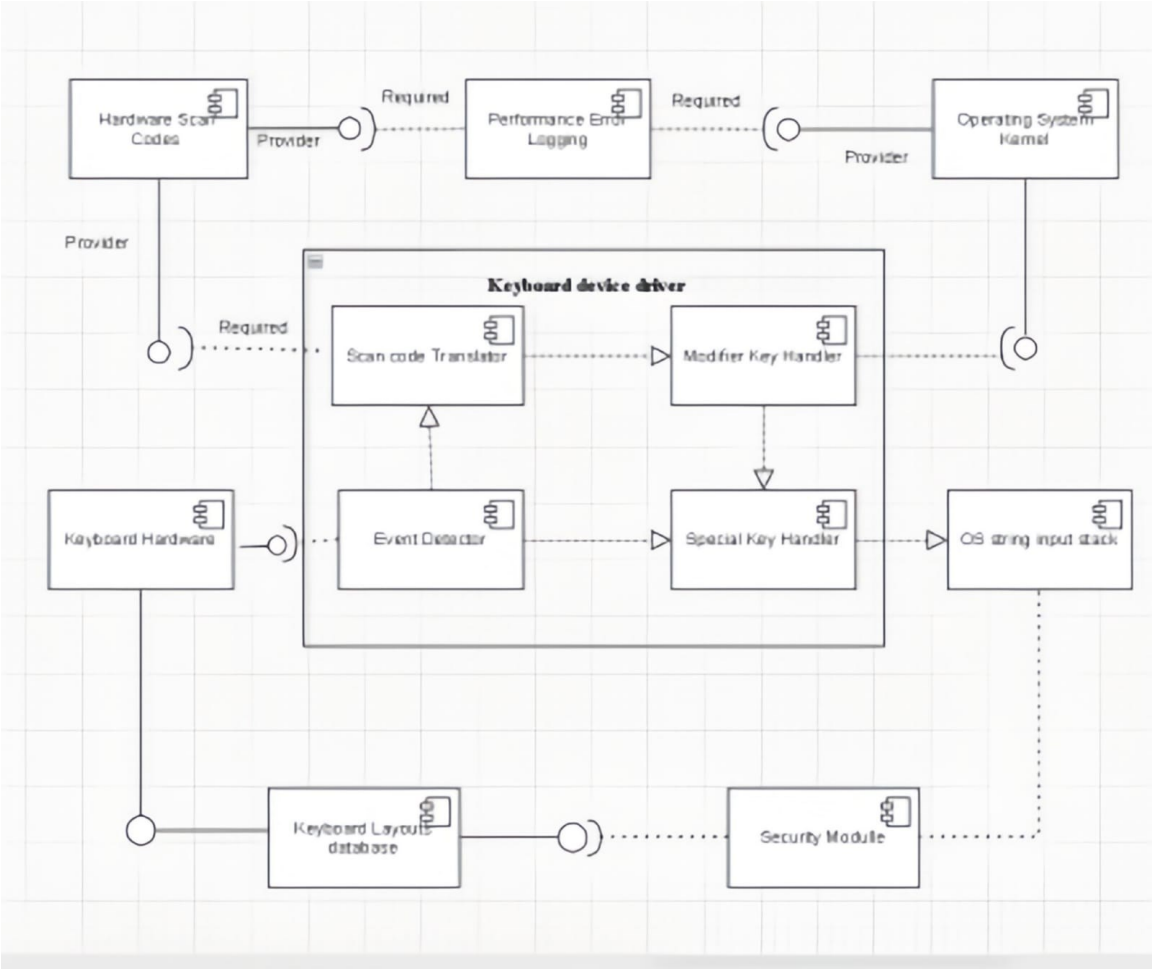

Figure 1keyboard device driver component diagram

Introduction

This report explains the working phases of the keyboard device driver based on the given component diagram. Each phase represents a logical step in how raw key presses from the keyboard hardware are converted into meaningful input for the operating system and applications.

Functional and Non-Functional requirements

Phase 1: Keyboard Hardware

This phase represents the physical keyboard. When a user presses or releases a key, the keyboard generates an electrical signal corresponding to that action. This signal is sent to the computer in the form of raw scan codes.

Phase 2: Event Detector

The event detector receives raw signals from the keyboard hardware. Its role is to detect key events such as key press and key release. Once an event is detected, it forwards the scan code to the next component for processing.

Phase 3: Scan Code Translator

In this phase, raw scan codes are translated into standardized key codes. This translation makes the input independent of the keyboard's hardware layout and prepares it for logical interpretation by the operating system.

Phase 4: Modifier Key Handler

The modifier key handler processes special modifier keys such as Shift, Ctrl, Alt, and Caps Lock. It updates the current keyboard state and modifies key codes accordingly before passing them forward.

Phase 5: Special Key Handler

This phase handles special-purpose keys like function keys (F1–F12), arrow keys, and system control keys. These keys may trigger specific system actions instead of producing character input.

Phase 6: OS String Input Stack

The OS string input stack receives fully processed key information. At this stage, characters and commands are delivered to the operating system and then to user applications such as text editors or browsers.

Conclusion

The diagram clearly shows a step-by-step flow from physical key press to meaningful system input. Each phase plays a crucial role in ensuring accurate, efficient, and consistent keyboard input handling.